Nour EL BACHARI



Dual-degree Engineer from Centrale Lyon and Arts et Métiers, specialised in robotics and artificial intelligence. Passionate about technological innovation, I aim to apply my expertise to the development of the robots of tomorrow.

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EXPERIENCE

R&D Engineering Intern — Robotics & Physical AI

Accenture

04/26 – 09/26 Paris

- *Grasping & Manipulation (Production 3, Franka)* : training and simulation-based validation of vision-based pick-and-place policies, with sim-to-real transfer.
- *Robot Interaction (Mirokai, Enchanted Tools)* : design and development of a multi-agent, multimodal, platform-agnostic architecture orchestrating human-robot interaction in real time.
- *Autonomous Visual Inspection (Spot, Boston Dynamics)* : design and deployment of an embedded visual perception system on a legged robot in an industrial environment.

R&D Engineering Intern — Industrial Automation & Robotics

Safran

06/25 – 08/25 Casablanca

- Integration study for autonomous mobile supply robots (AMR) in an industrial environment.
- Design and implementation of an industrial IoT platform for real-time tool tracking, including supervision and automated replenishment.

President

Robotics Club — Arts et Métiers

2024 – 2025 Meknès

- Team management and supervision of robotics projects.
- Organisation of robotics competitions within the school.
- Winner of several national robotics competitions.

PROJECTS

Usine 5.0 — Industrial Project

Arts et Métiers

2024 – 2025

- Design and implementation of an industrial bench with an augmented reality-assisted supervision and maintenance system.
- Design and prototyping of a robotic system comprising a robotic arm handling tool retrieval from stock, coupled with an AMR ensuring their autonomous delivery to the operator workstation.
- Development of the IoT and communication infrastructure interconnecting the entire system.

EDUCATION

Final Year Enginnering — Exchange Program

Centrale Lyon

2025 – 2026 Lyon

Track: Intelligent Transport
Specialisation: Entrepreneur & Intrapreneur Engineer

Master's in Electronics, Electrical Energy & Automation

Centrale Lyon

2025 – 2026 Lyon

Pathway: Electronics and Embedded Systems

Engineering Degree

Arts et Métiers

2021 – 2025 Meknès

Field: Electromechanical Engineering, Control and Industrial Management

SKILLS

Artificial Intelligence :
PyTorch, OpenCv, MediaPipe, CUDA, ONNX, CNN, ViT, VLM, VLA, Imitation Learning, Reinforcement Learning

Robotics :
ROS2, Nav2, MoveIt2, Isaac Sim, Isaac Lab, Modelling & Control, Kinematics & Dynamics

Embedded Systems & Mechatronics :
NVIDIA Jetson, Raspberry Pi, STM32, ESP32, Electronic Prototyping, 3D Printing, Communication Protocols (OPC-UA/MQTT)

Programming & DevOps :
Python, C/C++, SQL, Matlab/Simulink, Bash, Git, Docker, Linux

SOFT SKILLS

Autonomy, Analytical mindset, Scientific rigor, Creativity, Adaptability, Teamwork and Active listening.

LANGUAGES

- **French** : Native
- **English** : Fluent

INTERESTS

Robotics, Piano, Formula 1, Science and History.